

Raod accident SQL Queries

A.KPI's

1. CY Casualties

```
UPDATE
    road_accident
SET
    accident_date = STR_TO_DATE(accident_date, '%d-%m-%Y');

select
    sum(number_of_casualties) as CY_Casualties
from
    road_accident
where
    year(accident_date) = '2022'
```

CY_Casualties
195737

2. CY accident

```
select
    count(distinct(accident_index)) as CY_Accident
from
    road_accident
where
    year(accident_date) = '2022'
```

CY_Accident
144419

3. CY fatal Casualties

```
select
    sum(number_of_casualties)
from
    road_accident
where
    year(accident_date) = '2022'
    and accident_severity = 'Fatal'
```

CY_Fatal_casualties
2855

4. CY serious Casualties

```
select
    sum(number_of_casualties) as CY_Serious_casualties
from
    road_accident
where
    year(accident_date)= '2022'
    and accident_severity = 'Serious'
```

CY_Serious_casualties
27045

5. CY slight Casualties

```
select
    sum(number_of_casualties) as CY_Slight_casualties
from
    road_accident
where
    year(accident_date)= '2022'
    and accident_severity = 'Slight'
```

CY_Slight_casualties
165837

B. Granular Requirement

1. CY casualties by vehicle type

```
select
    case
        when vehicle_type in ('Agricultural vehicle') then 'Agricultural'
        when vehicle_type in ('Car', 'Taxi/Private hire car') then 'Cars'
        when vehicle_type in ('Motorcycle over 500cc', 'Motorcycle 125cc and
under', 'Motorcycle 50cc and under', 'Motorcycle over 125cc and up to 500cc') then
'Motorcycle'
        when vehicle_type in ('Pedal cycle') then 'Bike'
        when vehicle_type in ('Van / Goods 3.5 tonnes mgw or under', 'Goods over
3.5t. and under 7.5t', 'Goods 7.5 tonnes mgw and over') then 'Van'
        when vehicle_type in ('Bus or coach (17 or more pass seats)', 'Minibus (8 -
16 passenger seats)') then 'Bus'
        else 'Other vehicle'
```

```

        end as vehicle_group,
        sum(number_of_casualties) as CY_Casualties
from
    road_accident
where
    year(incident_date)='2022'
group by
    vehicle_group

```

vehicle_group	CY_Casualties
Cars	155804
Motorcycle	15579
Van	15905
Bus	6573
Other vehicle	1446
Agricultural	399
Bike	31

C. Chart's Requirement

1. CY vs PY casualties monthly trend

```

select
    monthname(incident_date) as month,
    sum(case when year(incident_date)='2022' then number_of_casualties else 0 end)
as CY_casualties,
    sum(case when year(incident_date)='2021' then number_of_casualties else 0 end)
as PY_casualties,
    sum(case when year(incident_date)='2022' then number_of_casualties else 0
end)-sum(case when year(incident_date)='2021' then number_of_casualties else 0 end)
as difference
from
    road_accident
group by
    month

```

Or if not working, instead of group by month put :
group by
 monthname(incident_date) as month

month	CY_casualties	PY_casualties	difference
January	13163	18173	-5010
February	14804	14648	156
March	16575	17815	-1240
April	15767	17335	-1568
May	16775	18852	-2077
June	17230	18728	-1498
August	16796	18797	-2001
July	17201	19682	-2481
September	17500	18456	-956
October	18287	20109	-1822
November	18439	20975	-2536
December	13200	18576	-5376

2. Casualties by road type

```

select
    road_type,
    sum(number_of_casualties) as CY_casualties
from
    road_accident
where
    year(accident_date) = '2022'
group by
    road_type
order by
    sum(number_of_casualties) desc

```

road_type	CY_casualties
Single carriageway	144653
Dual carriageway	31912
Roundabout	12683
One way street	3499
Slip road	2990

3. Casualties by urban/rural

```

select
    urban_or_rural_area,
    sum(case when year(accident_date) = '2022' then (number_of_casualties)
    else 0 end)/(select sum(number_of_casualties) from road_accident where
    year(accident_date)='2022')*100 as PCT_CY_casualties,

```

```

sum(case when year(accident_date) = '2021' then (number_of_casualties)
else 0 end)/(select sum(number_of_casualties) from road_accident where
year(accident_date)='2021')*100 as PCT_PY_casualties
from
road_accident
group by
urban_or_rural_area

```

urban_or_rural_area	PCT_CY_casualties	PCT_PY_casualties
Urban	61.9459	60.5966
Rural	38.0541	39.4034

4. Casualties by light conditions

```

select
case
when light_conditions in ('Darkness - lights lit', 'Darkness - lighting
unknown', 'Darkness - lights unlit', 'Darkness - no lighting') then 'Night'
else 'Day'
end as group_light,
sum(case when year(accident_date) = '2022' then (number_of_casualties)
else 0 end)/(select sum(number_of_casualties) from road_accident where
year(accident_date)='2022')*100 as PCT_CY_casualties,
sum(case when year(accident_date) = '2021' then (number_of_casualties)
else 0 end)/(select sum(number_of_casualties) from road_accident where
year(accident_date)='2021')*100 as PCT_PY_casualties
from
road_accident
group by
group_light

```

Or if not working, instead of group by group_light put :

```

group by
case
when light_conditions in ('Darkness - lights lit', 'Darkness - lighting
unknown', 'Darkness - lights unlit', 'Darkness - no lighting') then 'Night'
else 'Day'
end

```

group_light	PCT_CY_casualties	PCT_PY_casualties	
Day	73.8435	72.2156	
Night	26.1565	27.7844	

5. Top 10 locations by No of casualties

```

select
    local_authority,
    sum(case when year(accident_date) = '2022' then (number_of_casualties)
else 0 end) as CY_casualties,
    sum(case when year(accident_date) = '2021' then (number_of_casualties)
else 0 end) as PY_casualties
from
    road_accident
group by
    local_authority
order by
    CY_casualties desc
limit 10

```

local_authority	CY_casualties	PY_casualties	
Birmingham	4092	4519	
Leeds	2764	3057	
Cornwall	2092	1728	
Bradford	2089	2342	
Liverpool	2077	1975	
Manchester	1962	2404	
Sheffield	1764	1973	
County Durham	1708	1587	
Cheshire East	1656	1511	
Kirklees	1614	1698	